

Ay 5 Inferencia

P1) Solucionada Ay 4 P4

P2) $X_1, \dots, X_n \sim \text{Rayleigh}(\sigma)$ $\sigma > 0$

$$f(x|\sigma) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) I_{(0,\infty)}(x)$$

a) Condiciones de regularidad: 1) $X \in [0, \infty)$ ✓

$$2) \theta = (\sigma, \sigma) \quad \frac{-2}{\sigma} + \frac{1}{\sigma^3} E(X^2) = \frac{-2}{\sigma} + \frac{1}{\sigma^3}$$

$$E(X) = \int_0^\infty x \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx$$

$$3) \frac{\partial f}{\partial \sigma} \text{ existe } \int \frac{\partial f}{\partial \sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) + \frac{x}{\sigma^2} \frac{x^2}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

$$4) \frac{\partial}{\partial \sigma} \int f = \int \frac{\partial f}{\partial \sigma} \text{ es calculado se puede integrar el límite}$$

$$\frac{\partial}{\partial \sigma} \int f = 0$$

$$\int \frac{\partial f}{\partial \sigma} = \int \left\{ -\frac{2x}{\sigma^3} \exp\left(-\frac{x^2}{2\sigma^2}\right) + \frac{x^3}{\sigma^4} \exp\left(-\frac{x^2}{2\sigma^2}\right) \right\}$$

$$= -\frac{2}{\sigma} \int f + \frac{1}{\sigma^3} E(X^3)$$

$$= -\frac{2}{\sigma} + \frac{1}{\sigma^3} \cdot 2\sigma^2 = -\frac{2}{\sigma} + \frac{2}{\sigma} = 0$$

$$E(X^2) = \int_0^\infty x^2 \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx$$

$$5) \frac{\partial f}{\partial \sigma^k} < \infty \rightarrow \text{se puede}$$

$$6) E\left[\left(\frac{\partial}{\partial \sigma} \log f(X|\sigma)\right)^2\right]$$

$$E\left[\left(\frac{\partial}{\partial \sigma} \log(X) - \log(\sigma^2) + \frac{-X^2}{\sigma^2}\right)^2\right]$$

$$E\left[\left(-\frac{2X}{\sigma^2} + \frac{2X^2}{\sigma^3}\right)^2\right] < \infty$$

$$b) f(\vec{x}|\theta) = \frac{1}{h(\vec{x})} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2\right\}$$

Como es una familia exponencial $\rightarrow T(X) = \sum_{i=1}^n x_i^2$ es suficiente mínima completa

$$\begin{aligned} c) E(T^2 X_i^2) &= \sum E(X_i^4) \\ &= \sum \int \frac{\partial}{\partial \theta} \log(p(\theta)) \\ &= \sum \int \frac{\partial}{\partial \theta} \left(-\frac{1}{2\sigma^2} \sum x_i^2\right) \\ &= \sum \int \frac{\partial}{\partial \theta} \left(-\frac{1}{2\sigma^2}\right) \\ &= \sum \int \frac{\partial}{\partial \theta} \left(-\frac{1}{2\sigma^2}\right) \\ &= \sum \int \frac{\partial}{\partial \theta} \left(-\frac{1}{2\sigma^2}\right) \\ &= \sum \int \frac{\partial}{\partial \theta} \left(-\frac{1}{2\sigma^2}\right) \end{aligned}$$

$$\begin{aligned} E\left(\frac{1}{2\sigma^2} \sum x_i^2\right) &= -2 \frac{\partial}{\partial \theta} \log(\sigma) \\ E\left(\frac{x_i^2}{\sigma^2}\right) &= \frac{1}{\sigma^2} E(X_i^2) = \frac{2}{\sigma^2} \\ E(X_i^2) &= 2\sigma^2 \end{aligned}$$

Def: Condiciones de Regularidad

1) X espacio muestral no depende de θ

2) θ espacio paramétrico es un intervalo abierto de $\mathbb{R}$

3) $\frac{\partial f}{\partial \theta}$ existe y es finita

$$4) \frac{\partial}{\partial \theta} \int f(x|\theta) dx = \int \frac{\partial f(x|\theta)}{\partial \theta} dx \quad \forall \theta$$

5) $\forall T(X)$ es aditiva

$$\frac{\partial^k}{\partial \theta^k} \int T(X) f(x|\theta) dx = \int T(X) \frac{\partial^k f}{\partial \theta^k}$$

$$6) E\left[\left(\frac{\partial}{\partial \theta} \log f(X|\theta)\right)^2\right] < \infty \quad \forall \theta$$

Def Verosimilitud

$$L(\sigma) = f(\vec{X}|\sigma)$$

$$\ell(\sigma) = \ln(L(\sigma)) = \sum_{i=1}^n \log(f(X_i|\sigma))$$

Def: Función score

$$U(\sigma|x) = \frac{\partial \ell(\sigma)}{\partial \sigma}$$

Teorema: Bajo regularidad

$$E(U(\sigma|x)) = 0 \quad \forall \sigma$$

Def: Información de Fisher

$$I_X(\sigma) = E[U^2(\sigma|x)] = E\left[\left(\frac{\partial}{\partial \sigma} \log f(X|\sigma)\right)^2\right]$$

Teorema

$$\begin{aligned} I_X(\sigma) &= \text{Var}(U(\sigma|x)) \\ &= -E\left[\frac{\partial^2}{\partial \sigma^2} \log f(X|\sigma)\right] \end{aligned}$$

Teorema X y Y independientes

$$I_{(X,Y)}(\sigma) = I_X(\sigma) + I_Y(\sigma)$$

$$\text{Var}\left(\sum_{i=1}^n X_i^2\right) = \sum_{i=1}^n \text{Var}(X_i^2) = n \cdot 4\sigma^4$$

$$\text{Var}\left(\frac{\partial}{\partial \sigma} \frac{1}{2\sigma^2} X_i^2\right) = -\frac{\partial^2 \ell(\sigma^2)}{\partial \sigma^2} - \mathbb{E}\left[\left(\frac{\partial}{\partial \sigma^2} \frac{1}{2\sigma^2} X_i^2\right)^2\right]$$

$$\text{Var}\left(\frac{1}{\sigma^3} X_i^2\right) = -\frac{2}{\sigma^2} - \mathbb{E}\left[\frac{3}{\sigma^4} X_i^2\right]$$

$$= -\frac{2}{\sigma^2} + \frac{3}{\sigma^4} \mathbb{E}(X_i^2)$$

$$\sigma^{-6} \text{Var}(X_i^2) = \frac{-2}{\sigma^2} + \frac{3 \cdot 2\sigma^2}{\sigma^4}$$

$$\sigma^{-6} \text{Var}(X_i^2) = \frac{4}{\sigma^2} \quad | \cdot \sigma^6$$

$$\text{Var}(X_i^2) = \frac{4\sigma^4}{1} = \mathbb{E}(X_i^4) - \mathbb{E}(X_i^2)^2$$

$$= \sigma^4 \cdot 3 - \sigma^4 = 2\sigma^4$$

d) Función Score

$$U(\sigma|x) = \frac{\partial \ell(\sigma)}{\partial \sigma} = \frac{\partial}{\partial \sigma} \left\{ n \log(x) - 2n \log(\sigma) - \frac{1}{2\sigma^2} \sum X_i^2 \right\}$$

$$U(\sigma|x) = -\frac{2n}{\sigma} + \frac{1}{\sigma^3} \sum X_i^2$$

$$\mathbb{E}(U(\sigma|x)) = \mathbb{E}\left(-\frac{2n}{\sigma} + \frac{1}{\sigma^3} \sum X_i^2\right)$$

$$= -\frac{2n}{\sigma} + \frac{1}{\sigma^3} \mathbb{E}\left(\sum X_i^2\right) = -\frac{2n}{\sigma} + \frac{1}{\sigma^3} 2n\sigma^2 = 0 \quad (\text{Teorema})$$

op1
Teorema

$$I_X(\sigma) = \text{Var}(U(\sigma|x)) = \mathbb{E}(U(\sigma|x)^2) = \mathbb{E}$$

$$\text{Var}\left(-\frac{2n}{\sigma} + \frac{1}{\sigma^3} \sum X_i^2\right)$$

$$\frac{1}{\sigma^6} \text{Var}\left(\sum X_i^2\right) = \frac{1}{\sigma^6} \cdot n \cdot 4\sigma^4 = \frac{4n}{\sigma^2}$$

op2

$$I_X(\sigma) = \text{Var}(U(\sigma|x))$$

$$= -\mathbb{E}\left[\left(\frac{\partial}{\partial \sigma} \log f\right)^2\right]$$

$$= -\mathbb{E}\left[\frac{\partial}{\partial \sigma} U(\sigma|x)\right]$$

$$= -\mathbb{E}\left(\frac{2n}{\sigma^2} - \frac{3}{\sigma^4} \sum X_i^2\right)$$

$$= -\frac{2n}{\sigma^2} + \frac{3}{\sigma^4} \mathbb{E}\left(\sum X_i^2\right)$$

$$= -\frac{2n}{\sigma^2} + \frac{3}{\sigma^4} \cdot 2n\sigma^2$$

$$= \frac{4n}{\sigma^2}$$

e) Linfo 1.4.4.1.1 op1 y op2

f) $I_X(\sigma) = I_{T(X)}(\theta)$ ya que $T(X)$ es suficiente

g) $f(x|\sigma) = X \cdot \sigma^{-2n} \exp\left[-\frac{1}{2\sigma^2} \sum X_i^2\right]$ $\sigma = \frac{1}{2\sigma^2} \rightarrow \sigma^2 = \frac{1}{2\sigma^2} \rightarrow \sigma^2 = \sigma^2 2^n$

$$= X \cdot \sigma^{-2n} 2^n \exp\left[-\sigma \sum X_i^2\right]$$

$$\ell(\theta) = n \log(x) + n \log(\sigma) + n \log(2) - \sigma \sum X_i^2$$

$$\frac{\partial \ell(\theta)}{\partial \sigma} = \frac{n}{\sigma} - \sum X_i^2 \rightarrow \frac{\partial^2 \ell(\theta)}{\partial \sigma^2} = -\frac{n}{\sigma^2}$$

$$I_X(\theta) = -\mathbb{E}\left[\frac{\partial^2 \ell(\theta)}{\partial \sigma^2}\right] = -\mathbb{E}\left(-\frac{n}{\sigma^2}\right) = \frac{n}{\sigma^2} = 4n\sigma^4$$

op2

$$I_X(\theta) = I_X(2\sigma)^{-1/2} \cdot \left(\frac{1}{2\sigma}\right)^2 = \frac{n}{\sigma^2}$$